

Table 1: CPU vs. GPU (standard) vs. GPU + Mooncake Connector — Qwen3-8B benchmark comparison
50 prompts/run, output.len=128. **GPU+Mooncake**: 2×V100S, 1P1D disaggregated, up to 32 768 tokens. **GPU standard**: 1×V100S, no Mooncake, fails at 32 768 tokens. **CPU**: vLLM CPU backend, stopped at 1 024 tokens.

Prompt	Backend	Duration (s)	Out tok/s	Total tok/s	TTFT mean (ms)	TTFT med. (ms)	TTFT P99 (ms)	TPOT mean (ms)	TPOT med.
	GPU + Mooncake	79.27	80.73	161.47	169.40	108.15	1672.14	11.15	—
	GPU standard	145.13	44.10	90.95	134.32	92.15	1169.48	21.80	21.79
	CPU	1345.05	4.76	9.81	7358.65	7315.55	8457.82	153.87	153.86
	GPU + Mooncake	78.99	81.02	243.06	151.38	150.54	174.96	11.25	—
	GPU standard	144.91	44.17	135.26	105.60	105.53	108.33	21.99	21.98
	CPU	1688.11	3.79	11.61	14 148.40	14 147.98	14 167.38	154.44	154.42
	GPU + Mooncake	84.07	76.13	380.63	231.72	230.15	252.91	11.41	—
	GPU standard	148.87	42.99	217.63	149.08	149.17	151.73	22.27	22.27
	CPU	2378.00	2.69	13.62	27 925.46	27 925.10	27 970.19	154.60	154.58
	GPU + Mooncake	98.42	65.03	585.27	477.35	476.05	515.38	11.74	—
	GPU standard	154.47	41.43	375.47	259.04	259.01	260.86	22.28	22.28
	CPU	3967.91	1.61	14.62	59 531.51	59 533.08	59 599.94	156.11	156.11
2048	GPU + Mooncake	135.55	47.22	802.68	1139.60	1134.08	1287.37	12.37	—
	GPU standard	166.74	38.38	654.91	484.95	485.19	488.62	22.44	22.44
4096	GPU + Mooncake	239.59	26.71	881.49	3059.94	3055.30	3118.31	13.63	—
	GPU standard	194.25	32.95	1089.30	987.57	988.17	997.36	22.81	22.81
8192	GPU + Mooncake	572.27	11.18	726.92	9393.10	9396.74	9433.55	16.16	—
	GPU standard	268.75	23.81	1549.40	2263.41	2264.23	2277.50	24.50	24.50
16384	GPU + Mooncake	1731.45	3.70	476.83	31 937.29	31 935.59	31 983.16	21.19	—
	GPU standard	457.46	13.99	1805.62	5771.98	5774.19	5846.68	26.59	26.59
32768	GPU + Mooncake	6025.21	1.06	272.99	116 533.66	116 533.19	116 590.04	31.26	—
	GPU standard	FAILED — Bad Request (context window exceeded)							

■ GPU + Mooncake: 2×V100S, disaggregated prefill/decode (1P1D), full range 128–32 768 tokens. ■ GPU standard: 1×V100S, no disaggregation, context limit hit at 32 768 tokens. ■ CPU: vLLM CPU backend, stopped at 1 024 tokens (too slow beyond that). — = metric not recorded for this run.